

A Pilot Study on People's Views of Gratitude Practices and Reactions to Expressing Gratitude in an Online Community

Bridget Ho*
bridgetho11@gmail.com
UC Santa Cruz
Santa Cruz, USA

Reina Itakura
UC Santa Cruz
Santa Cruz, USA
reitakur@ucsc.edu

Kehua Lei*
UC Santa Cruz
Santa Cruz, USA
klei4@ucsc.edu

Kathleen Lum
UC Santa Cruz
Santa Cruz, USA
kathleenlum68@gmail.com

Jonathan He
UC Santa Cruz
Santa Cruz, USA
jhe262@ucsc.edu

David Lee
UC Santa Cruz
Santa Cruz, USA
dlee105@ucsc.edu

ABSTRACT

Online communities play an increasingly important role in forming and maintaining relationships, but have also been criticized for their negative effects such as decreased mental well-being, reduced in-person social connections, and the spread of misinformation. This works-in-progress paper describes formative work towards designing online communities centered on gratitude, in which we elicited views on gratitude and on expressing gratitude within online communities. We found that people have unique views of and contexts for gratitude and discussed how the unique views and contexts relate to concerns they have around expressing gratitude in an online community. We describe plans for a follow-up study to better understand people's views on expressing gratitude in an online community through the use of Gratitude, a simple online communal gratitude platform where users interact in ways similar to many online communities, through creating short posts (e.g., notes of gratitude) that other members can browse, react to, and comment on.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Gratitude, online communities, prosocial behavior

ACM Reference Format:

Bridget Ho, Kehua Lei, Jonathan He, Reina Itakura, Kathleen Lum, and David Lee. 2023. A Pilot Study on People's Views of Gratitude Practices and Reactions to Expressing Gratitude in an Online Community. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, Article 37, 7 pages. <https://doi.org/10.1145/3584931.3607000>

*Both authors contributed equally to this research.



This work is licensed under a [Creative Commons Attribution-NonCommercial-NoDerivs International 4.0 License](https://creativecommons.org/licenses/by-nc-nd/4.0/).

CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA
© 2023 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0129-0/23/10.
<https://doi.org/10.1145/3584931.3607000>

1 INTRODUCTION

Online communities are becoming more important as people make new connections, stay in touch with friends, and socialize over long distances [63]. However, despite many demonstrated benefits [18, 89], social media sites have also been heavily criticized for decreasing real-life social communication, increasing depression [38, 50], facilitating misinformation [65], and breeding toxic anti-social interactions [66]. Prior research has demonstrated that grateful reflection is associated with improved well-being and prosocial behaviors that build community [5, 23, 25, 33]. In the context of online communities, studies have found that both expressing and receiving gratitude increased a sense of relatedness within online volunteering communities, predicting member engagement and well-being [103]. Moreover, a growing body of work on communal gratitude has led to the development of a “gratitude cycle” framework, which models how expressing gratitude to members of a community who perform kind acts creates a positive cycle of altruistic behavior and gratitude expression [55].

These observations lead us to ask: How might we best design prosocial online communities centered on gratitude? What insights might this teach us about how a focus on positive activities — such as grateful reflections — may create prosocial online communities? We believe that shared, public expressions of gratitude may foster positive connections with other members in the online community, yet we know very little about the impact that these actions might have. How might members feel about posting about gratitude publicly? How might this affect the way they use the online platform, given their experiences with other social media platforms?

In what follows, we discuss related literature on online prosocial and antisocial behaviors, as well as gratitude, (Sections 2) and then describe early formative interviews in which we elicited views on gratitude and views on expressing gratitude in an online community context. We present the unique views of and contexts for gratitude that people expressed and how these related to concerns they had about expressing gratitude in an online community (Sections 3). We then discuss promising directions we see for design (Sections 4) followed by a description of a simple gratitude-centered online community that we developed and our plans for our full study that will: 1) expand on our early interviews and 2) use our simple platform to further elicit insights on the opportunities and challenges in designing an online community centered on gratitude (Sections 5). Our hope is that this study can provide some lessons for how to

design online communities supporting positive social connections while minimizing negative impact on members.

2 RELATED WORK

Prosocial and Antisocial Behaviors in Online Communities

The explosive growth of online communities has created intellectual, social, and cultural value for its members [80], as well as economic benefits for commercial brands [71]. Active participation and contribution of users are crucial for the success of online communities [46, 64]. However, this has also given rise to a significant amount of anti-social behaviors, including but not limited to trolling [8, 36, 74], harassment [32, 92, 93], and cyberbullying [4, 101]. These behaviors can have a negative impact on individuals' psychological well-being [7, 60, 68], leading to further antisocial behaviors [15, 16]. Current research on online antisocial behaviors focuses on understanding the motivations behind antisocial behaviors [53, 81, 85, 91] and exploring how to reduce them through multiple methods, such as moderation [11–13, 39, 45, 51], prediction [75], governance [27, 76, 105], and deep listening [48, 49].

A large body of research also focuses on prosocial behaviors such as altruism [28, 54, 88], reciprocity [6, 30], and gratitude [41, 47], which tend to be interrelated with each other [58]. Prosocial behaviors can help users gain social support from other community members [96], which can translate into real-world support [73]. In HCI literature, a rich vein of work have investigated how prosocial behaviors exist in specific community types, such as altruism in gaming communities [61], reciprocity in online health communities [100, 102], and social learning in social media [22, 56]. However, a few studies have explored how to promote prosocial behaviors in online communities through platform designs [2, 78]. In this paper, we are interested in how centering an online community on positive, value-driven activities such as gratitude might help in the design of prosocial online communities.

Gratitude, Mental Health, and Group Dynamics

Gratitude has been defined in various ways in the literature [24, 59, 72, 97]. Generally, gratitude is conceptualized as a positive emotion that leads to prosocial behaviors [23, 57]. A wide range of studies in psychology have demonstrated its benefits for individuals' mental health [17, 69, 90], such as reducing stress [52] and promoting happiness [44]. While most research has focused on the individual benefits of gratitude [43, 62, 95], there is a growing interest in exploring its social nature, particularly in the context of groups [87].

As gratitude has been recognized as an inherently social emotion [10, 84] that can help strengthen social bonds [1, 3], there has been a growing call to examine its social nature [104]. Studies have shown that gratitude in groups can amplify positive emotional effects [83] by encouraging individuals to pass on kindness to others [14]. Several concepts have been proposed to explain gratitude dynamics in groups, including group-context gratitude where individuals practice gratitude while aware of others doing the same [20], collective gratitude as a shared emotion among members of an organization [29], and joint gratitude as a collective action [19]. Recently, a "gratitude cycle" framework has been proposed to present how

gratitude is organically expressed and acknowledged within existing online communities [55]. Our study aims to build on and integrate this literature towards an understanding of how to design a gratitude-centered online community. We seek to build on the concept of group-context gratitude to characterize practices of gratitude in the context of online communities, consider how one might encourage shared gratitude in the community, and explore the role of gratitude as a collective action in the design.

Digital Tools to Promote Practices of Gratitude

Gratitude interventions are commonly used to promote gratitude [26], requiring participants to engage in regular gratitude activities to enhance their gratitude experience [21, 79]. Compared to prior paper-based gratitude intervention research [67], recent studies have focused on exploring digital gratitude intervention techniques through, for example, text messages [70], augmented reality [9], social media [58], and speech assistants [37] to support gratitude activities. Expressing gratitude through digital technology can bring happiness [40] and produce fewer negative emotional effects than face-to-face communication [82]. In the post-COVID-19 world, digital gratitude intervention techniques have been utilized to help colleagues in remote work environments build positive connections [86, 99].

In recent years, with the significantly increased usage of social media, a few researchers have started to explore the relationship between gratitude interventions and SNSs (social networking sites) [35]. Koay et al. first reported the positive effects of using social media in gratitude interventions on Instagram [42], while Sciarra et al. further investigated how grateful interactions on Facebook can be observed, learned and imitated to amplify the spread of gratitude [77]. Additionally, research has shown that simply witnessing expressions of gratitude can enhance the life satisfaction and happiness of the observers [94]. Although some research has investigated the possibility of digital interventions in facilitating grateful interactions on social networking sites, we are not aware of any existing framework that actively integrates gratitude practices into the development of SNSs. Considering the capacity of social media to facilitate emotional contagion [31, 98], we believe that social media sites have the potential to promote gratitude activities in online communities, and that this in turn, can help with our understanding of designing prosocial online communities.

3 EARLY REACTIONS TO EXPRESSING GRATITUDE IN AN ONLINE COMMUNITY

To inform the design of our full study, we conducted a brief pilot study in which we conducted 30-minute long semi-structured interviews to gather qualitative data from representative users on their current experiences with social media, online communities, and gratitude practices, and their views and expectations for communal gratitude practices. For this early study, we interviewed 3 participants recruited through convenience sampling and analyzed the interviews through affinity mapping to determine common themes and insights.

Analysis of these three early interviews helped elicit three themes that point to two key areas of consideration. Two of our themes indicated hesitation around the adoption of communal gratitude

platforms. One additional theme centered on the diverse definitions of and contexts for gratitude. We briefly present these themes along with illustrative quotes and discuss potential implications for design in the following section.

Posting publicly may foster performative gratitude and ego-driven motivations

We discovered that some participants were worried about posting to boost their egos, rather than to communicate value or genuine gratitude towards others. Sharing appreciative messages in a community may feel performative, when compared to directly messaging a recipient privately.

- P1: "I think one thing that's scary about social media is that, like, you don't know if you're doing things to post for yourself, or if you're doing it for your audience."
- P2: "If you want to hear the uncharitable interpretation, the uncharitable interpretation is to flex on people."

Questioning the necessity or value of public expressions of gratitude

Beyond initial concerns about possible self-serving purposes of posting publicly, all three participants questioned the need to share their gratitude practice with others or to express appreciation to a recipient publicly. Grateful reflection can be a private practice for some, and it may not seem important or necessary to share those feelings with an audience.

- P2: "I think that it is kind of like a personal thing, like, if you are privately grateful ... I don't really see a reason why you have to share this or make it external."
- P3: "I think it's not necessary to share the gratitude practice with others."

Reactions reflected diverse definitions of, and contexts for, gratitude

The discussion around perceived authenticity of sharing gratitude messages publicly versus privately illustrates the importance of understanding how different people perceive gratitude practices, and what contexts they are more or less comfortable with. We discovered that participants may view the emotion of gratitude primarily as internal, external, or a combination of the two, which impacts their comfort with sharing expressions of gratitude within a community. Internal expressions of gratitude are classified as emotions or mindsets that are self-contained, whereas external expressions of gratitude have both a sender / writer and at least one recipient that is directly addressed.

- P1: "It's a mindset ... like training yourself to be mindful and considerate and grateful for things that you have."
- P2: "There's expression to yourself, and there's expression to other people, one being internal, one being external."
- P3: "I think gratitude is just the ability to express appreciation to others."

In addition, the context provided by the communities matters. Existing relationships and value systems may impact how a public message of gratitude is perceived or accepted. Public posts of gratitude towards others may be more acceptable when the relationship

between the writer and recipient is not especially personal, or if the recipient values public displays of appreciation.

- P2: "In a professional setting, I don't have a personal relationship to these people, and I do hope they get some sort of external award."
- P1: "Actually, I think it also depends on [their] love language, because I know some people are more receptive to public shows of affection."

4 DISCUSSION

Our early findings suggest that variations in how people define gratitude and the contexts within which they are expressing gratitude influence their desire to post publicly and the value they derive from an environment for communal gratitude. We see this as a promising direction for further exploration.

For example, what are the diverse goals and motivations for expressing gratitude and how should one best support them in a platform for communal gratitude? We saw at least a few different goals expressed in our small set of participants: growing an internal quality of being grateful, expressing gratitude for a person, and praising someone publicly. In the first two cases, it is clear that one has to be very careful about how public expressions of gratitude might create extrinsic motivations that crowd out intrinsic ones [34], and this is likely true for any online community that aims to center positive, value-driven activities. How might certain environments affect the decision to post gratitude messages in a community? Can public expressions of gratitude foster a collaborative feeling of sharing in, supporting, and receiving support for practices of gratitude rather than a performative one? What motivates individuals to post with the intention of impressing others versus truly appreciating them? What practices of gratitude should remain private, and are there ways that even private expressions of gratitude can contribute to communal aspects of gratitude?

A related consideration is the presence or absence of a recipient when expressing gratitude. Gratitude, when defined as a state of thankfulness independent of interpersonal relationships, has been researched using interventions such as journaling and linked to enhanced well-being. On the other hand, However, gratitude messages directed at specific recipients in response to interpersonal events may differ in format. Specifically, how do expressions without a recipient differ from those that have one, and how can we support these interactions within an online community (and our platform)? How does the presence or absence of a recipient affect intrinsic and extrinsic motivations around gratitude?

Finally, what are the diverse relational contexts that exist for expressing gratitude (e.g., personal, professional, affinity-based) and how should we best support them in a platform for communal gratitude? Preliminary findings suggest that in closer, personal relationships, there may be a preference for direct communication of gratitude (through a direct message or in-person) as it feels more authentic and less performative. On the other hand, in professional relationships or more distant friendships, public posts that show appreciation or acknowledgement seem to be more widely accepted. What is the value that people derive from these different relationships and how can practices of gratitude strengthen those relationships? How might external value systems in existing

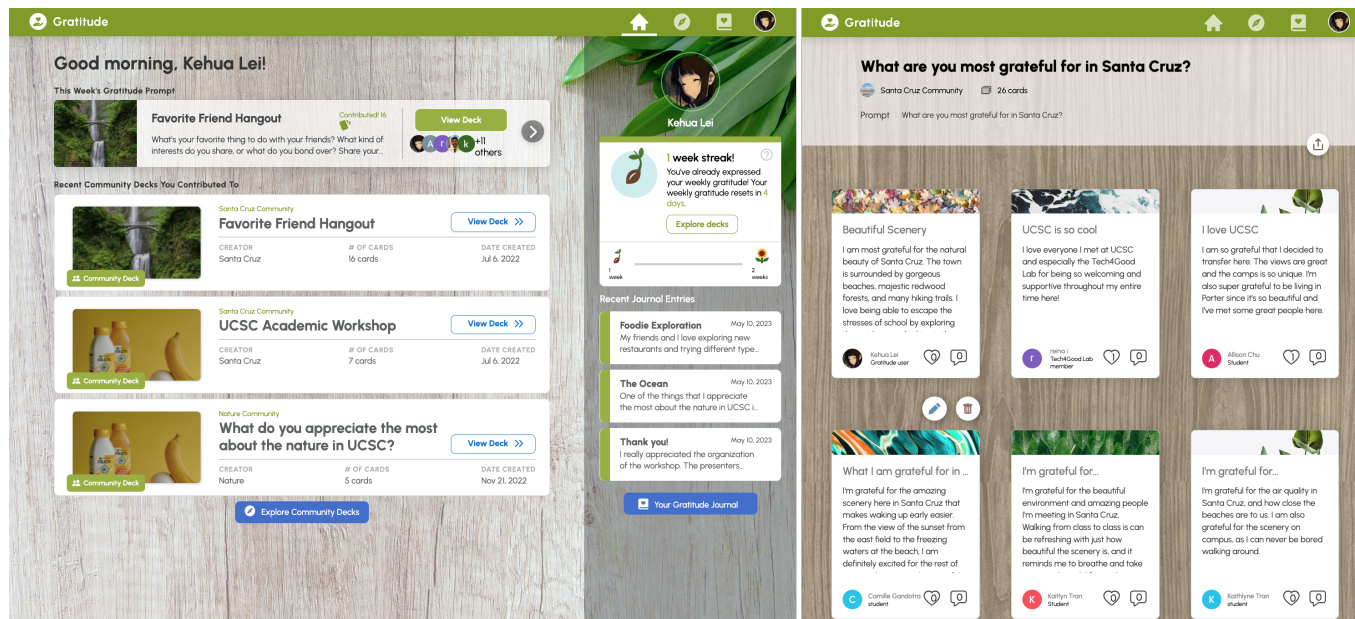


Figure 1: (Left) Users see weekly gratitude prompts, community decks they can contribute to, and journal entries of past gratitude notes written. (Right) Community decks consist of gratitude notes that users can browse and react to.

communities affect public expressions of gratitude, and how should communal practices of gratitude be designed in light of these?

As we explore these questions, we are also interested in understanding how the affordances of digital technologies and online communities might help one to thread the needle in supporting diverse goals and contexts through dynamics not possible in traditional ones. For example, suppose that having recipients complicates motivations for growing an internal state of thankfulness, but that having recipients can help foster a cycle of gratitude in the community. In that case, one might prompt people to write reflective notes of gratitude that do not have explicit recipients, but then extract out implicit audiences (e.g. teachers, park rangers, coffee baristas, etc.) from those notes to direct them to.

5 A SIMPLE GRATITUDE-CENTERED ONLINE COMMUNITY FOR STUDYING VIEWS ON COMMUNAL GRATITUDE

Besides improving the rigor of our semi-structured interviews and analysis, we also seek to more deeply elicit people's views on expressing gratitude in an online community by having participants use Gratitude, a simple platform that enables people to share and read expressions of gratitude from others in their communities.

Gratitude is designed to be analogous to the standard dynamics of online communities today, where people can make short posts to communities as well as browse and react to posts that others made, and is built using the Angular framework with Google Firebase for the backend database. We want to understand to what extent the standard dynamic was able to support communal gratitude and in what ways participants viewed it as limiting.

In Gratitude, a user is a member of one or more communities. The moderators of these communities can create gratitude prompts

that encourage reflection and expression of gratitude in the communities they are a part of. For the purpose of this study, we defined the prompts to elicit particular things we're interested in. Users see these gratitude prompts in the main dashboard view and can respond by writing a gratitude note. When they do so, their response is added to a 'community deck' corresponding to that prompt. All community members can navigate to these community decks to browse all responses for a prompt and react to them with emojis or comments. To encourage users in writing gratitude notes, the platform shows users their 'gratitude streak', the number of recent consecutive weeks during which they responded to a gratitude prompt. This streak is visualized in the form of a seed, which gradually blossoms into a flower. Finally, users can view past gratitude notes (their 'gratitude journal') and browse community decks.

For our study, we plan to recruit new participants from one community and interview them about their practices and views of gratitude, and then have them use Gratitude over a month-long period, after which we will analyze platform logs and interview them again to elicit further perspectives on their experience, their interactions with others, and their views on what they'd like to see in an online community for gratitude. We look forward to further exploring how these experiences and views relate to the unique ways that people view communal gratitude practices. Our hope is that this work will provide insights around how one might design platforms centered around gratitude specifically, as well as more broadly the design of prosocial online communities centered on other positive, value-driven activities.

ACKNOWLEDGMENTS

We thank to Allison Truong, Yukti Malhan, Camille Gandotra, Sai Kambampati, Allison Chu, Masumah Zahra, Angelica Tijerina,

Krupa Patel, Kevin Xie and Kamryn Callwood for contributing to the designs.

REFERENCES

- [1] Sara B Algoe, Shelly L Gable, and Natalya C Maisel. 2010. It's the little things: Everyday gratitude as a booster shot for romantic relationships. *Personal relationships* 17, 2 (2010), 217–233.
- [2] Ishwarya Ananthabhotla, Alexandra Rieger, Dan Greenberg, and Rosalind Picard. 2017. MIT community challenge: designing a platform to promote kindness and prosocial behavior. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 2352–2358.
- [3] Christina N Armenta, Megan M Fritz, Lisa C Walsh, and Sonja Lyubomirsky. 2020. Satisfied yet striving: Gratitude fosters life satisfaction and improvement motivation in youth. *Emotion* (2020).
- [4] Zahra Ashktorab and Jessica Vitak. 2016. Designing cyberbullying mitigation and prevention solutions through participatory design with teenagers. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 3895–3905.
- [5] Monica Y Bartlett and David DeSteno. 2006. Gratitude and prosocial behavior: Helping when it costs you. *Psychological science* 17, 4 (2006), 319–325.
- [6] Daniel Belanche, Luis V Casaló, Carlos Flavián, and Miguel Guinalíu. 2019. Reciprocity and commitment in online travel communities. *Industrial management & data systems* 119, 2 (2019), 397–411.
- [7] Tanya Beran and Qing Li. 2005. Cyber-harassment: A study of a new method for an old behavior. *Journal of educational computing research* 32, 3 (2005), 265.
- [8] Amy Binns. 2012. DON'T FEED THE TROLLS! Managing troublemakers in magazines' online communities. *Journalism practice* 6, 4 (2012), 547–562.
- [9] Benjamin Bunn, Wesley Colemere, Tristan Moe, Colin Rampton, and Amanda Lee Hughes. 2022. GoGratitude: Using Augmented Reality to Inspire Gratitude. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–6.
- [10] Natàlia Cantó-Milà. 2013. Gratitude—invisibly webbing society together. *Journal of classical sociology* 13, 1 (2013), 8–19.
- [11] Stevie Chancellor, Yannis Kalantidis, Jessica A Pater, Munmun De Choudhury, and David A Shamma. 2017. Multimodal classification of moderated online pro-eating disorder content. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 3213–3226.
- [12] Eshwar Chandrasekharan, Chaitrali Gandhi, Matthew Wortley Mustelier, and Eric Gilbert. 2019. Crossmod: A cross-community learning-based system to assist reddit moderators. *Proceedings of the ACM on human-computer interaction* 3, CSCW (2019), 1–30.
- [13] Eshwar Chandrasekharan and Eric Gilbert. 2019. Hybrid approaches to detect comments violating macro norms on reddit. *arXiv preprint arXiv:1904.03596* (2019).
- [14] Yen-Ping Chang, Yi-Cheng Lin, and Lung Hung Chen. 2012. Pay it forward: Gratitude in social networks. *Journal of Happiness Studies* 13 (2012), 761–781.
- [15] Justin Cheng, Michael Bernstein, Cristian Danescu-Niculescu-Mizil, and Jure Leskovec. 2017. Anyone can become a troll: Causes of trolling behavior in online discussions. In *Proceedings of the 2017 ACM conference on computer supported cooperative work and social computing*. 1217–1230.
- [16] Justin Cheng, Cristian Danescu-Niculescu-Mizil, and Jure Leskovec. 2015. Anti-social behavior in online discussion communities. In *Proceedings of the international aaai conference on web and social media*, Vol. 9. 61–70.
- [17] Sheung-Tak Cheng, Pui Ki Tsui, and John HM Lam. 2015. Improving mental health in health care practitioners: randomized controlled trial of a gratitude intervention. *Journal of consulting and clinical psychology* 83, 1 (2015), 177.
- [18] Jenna L Clark, Sara B Algoe, and Melanie C Green. 2018. Social network sites and well-being: The role of social connection. *Current Directions in Psychological Science* 27, 1 (2018), 32–37.
- [19] Joshua Cockayne and Gideon Salter. 2022. Liturgical gratitude to God. *Religions* 13, 9 (2022), 795.
- [20] Joshua Cockayne and Gideon Salter. 2023. Group Gratitude: A Taxonomy. *The Journal of Value Inquiry* (2023), 1–22.
- [21] Natalia Czyżowska and Ewa Gurba. 2022. Enhancing meaning in life and psychological well-being among a european cohort of young adults via a gratitude intervention. *Frontiers in Psychology* 12 (2022), 6263.
- [22] Dominic DiFranzo, Yoon Hyung Choi, Amanda Purington, Jessie G Taft, Janis Whitlock, and Natalya N Bazarova. 2019. Social media testdrive: Real-world social media education for the next generation. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–11.
- [23] B Eamons and Michael E McCullough. 2003. Counting blessings versus burdens: An experimental investigation of gratitude and subjective well-being in daily life. *Journal of Personality and Social Psychology* 84, 2 (2003), 377–389.
- [24] Robert A Emmons, Michael E McCullough, et al. 2004. *The psychology of gratitude*. Oxford University Press.
- [25] Robert A Emmons and Charles M Shelton. 2002. Gratitude and the science of positive psychology. *Handbook of positive psychology* 18 (2002), 459–471.
- [26] Robert A Emmons and Robin Stern. 2013. Gratitude as a psychotherapeutic intervention. *Journal of clinical psychology* 69, 8 (2013), 846–855.
- [27] Jenny Fan and Amy X Zhang. 2020. Digital juries: A civics-oriented approach to platform governance. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–14.
- [28] Sizheng Fan, Tian Min, Xiao Wu, and Wei Cai. 2023. Altruistic and Profit-oriented: Making Sense of Roles in Web3 Community from Airdrop Perspective. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [29] Ryan Fehr, Ashley Fulmer, Eli Awtrey, and Jared A Miller. 2017. The grateful workplace: A multilevel model of gratitude in organizations. *Academy of Management Review* 42, 2 (2017), 361–381.
- [30] Yuanyue Feng and Hua Jonathan Ye. 2016. Why do you return the favor in online knowledge communities? A study of the motivations of reciprocity. *Computers in Human Behavior* 63 (2016), 342–349.
- [31] Emilio Ferrara and Zeyao Yang. 2015. Measuring emotional contagion in social media. *PLoS one* 10, 11 (2015), e0142390.
- [32] Antigoni Founta, Constantinos Djouvas, Despoina Chatzakou, Ilias Leontiadis, Jeremy Blackburn, Gianluca Stringhini, Athena Vakali, Michael Sirivianos, and Nicolas Kourtellis. 2018. Large scale crowdsourcing and characterization of twitter abusive behavior. In *Proceedings of the international AAAI conference on web and social media*, Vol. 12.
- [33] Barbara L Fredrickson. 2004. Gratitude, like other positive emotions, broadens and builds. *The psychology of gratitude* 145 (2004), 166.
- [34] Bruno S Frey. 1994. How intrinsic motivation is crowded out and in. *Rationality and society* 6, 3 (1994), 334–352.
- [35] Ericka M Goerling. 2016. *Gratitude in the 21 st Century: An Experimental Examination of Gratitude Expressions on Facebook and Resultant Subjective Well-Being*. Ph.D. Dissertation. Fielding Graduate University.
- [36] Claire Hardaker. 2010. Trolling in asynchronous computer-mediated communication: From user discussions to academic definitions. (2010).
- [37] André Helgert and Carolin Straßmann. 2022. What are you Grateful for? Enhancing Gratitude Routines by Using Speech Assistants. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–7.
- [38] Melissa G Hunt, Rachel Marx, Courtney Lipson, and Jordyn Young. 2018. No more FOMO: Limiting social media decreases loneliness and depression. *Journal of Social and Clinical Psychology* 37, 10 (2018), 751–768.
- [39] Shagun Jhaver, Quan Ze Chen, Detlef Knauss, and Amy X Zhang. 2022. Designing Word Filter Tools for Creator-led Comment Moderation. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–21.
- [40] Alison Killen and Ann Macaskill. 2015. Using a gratitude intervention to enhance well-being in older adults. *Journal of happiness studies* 16 (2015), 947–964.
- [41] Yongsung Kim, Daishi Kato, Kazuo Kunieda, and Keiji Yamada. 2013. Preliminary user study for gratitude and reciprocity in a Q&A system. In *Proceedings of the 2013 conference on Computer supported cooperative work companion*. 169–174.
- [42] Siew-Hoong Koay, Ai-Tsin Ng, Shi-Ki Tham, and Chee-Seng Tan. 2020. Gratitude intervention on Instagram: An experimental study. *Psychological Studies* 65 (2020), 168–173.
- [43] Yu Komase, Kazuhiro Watanabe, Daisuke Hori, Kyosuke Nozawa, Yui Hidaka, Mako Iida, Kotaro Imamura, and Norito Kawakami. 2021. Effects of gratitude intervention on mental health and well-being among workers: A systematic review. *Journal of Occupational Health* 63, 1 (2021), e12290.
- [44] Feng Kong, Ke Ding, and Jingjing Zhao. 2015. The relationships among gratitude, self-esteem, social support and life satisfaction among undergraduate students. *Journal of Happiness Studies* 16, 2 (2015), 477–489.
- [45] Yubo Kou and Xinning Gui. 2021. Flag and flaggability in automated moderation: The case of reporting toxic behavior in an online game community. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [46] Robert E Kraut and Paul Resnick. 2012. *Building successful online communities: Evidence-based social design*. MIT Press.
- [47] Mary Jo Kreitzer, Susan Telke, Linda Hanson, Brent Leininger, and Roni Evans. 2019. Outcomes of a gratitude practice in an online community of caring. *The Journal of Alternative and Complementary Medicine* 25, 4 (2019), 385–391.
- [48] Travis Kriplean, Jonathan Morgan, Deen Freelon, Alan Borning, and Lance Bennett. 2012. Supporting reflective public thought with considerit. In *Proceedings of the ACM 2012 conference on Computer Supported Cooperative Work*. 265–274.
- [49] Travis Kriplean, Michael Toomim, Jonathan Morgan, Alan Borning, and Amy J Ko. 2012. Is this what you meant? Promoting listening on the web with reflect. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 1559–1568.
- [50] Daria J Kuss and Mark D Griffiths. 2011. Online social networking and addiction—a review of the psychological literature. *International journal of environmental research and public health* 8, 9 (2011), 3528–3552.
- [51] Vivian Lai, Samuel Carton, Rajat Bhatnagar, Q Vera Liao, Yunfeng Zhang, and Chenhao Tan. 2022. Human-ai collaboration via conditional delegation: A case

- study of content moderation. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [52] Ju-Yeon Lee, Seon-Young Kim, Kyung-Yeol Bae, Jae-Min Kim, Il-Seon Shin, Jin-Sang Yoon, and Sung-Wan Kim. 2018. The association of gratitude with perceived stress and burnout among male firefighters in Korea. *Personality and Individual Differences* 123 (2018), 205–208.
 - [53] So-Hyun Lee and Hee-Woong Kim. 2015. Why people post benevolent and malicious comments online. *Commun. ACM* 58, 11 (2015), 74–79.
 - [54] Will WK Ma and Albert Chan. 2014. Knowledge sharing and social media: Altruism, perceived online attachment motivation, and perceived online relationship commitment. *Computers in human behavior* 39 (2014), 51–58.
 - [55] Stephann Makri and Sophie Turner. 2020. “I can’t express my thanks enough”: The “gratitude cycle” in online communities. *Journal of the Association for Information Science and Technology* 71, 5 (2020), 503–515.
 - [56] Stefania Manca and Maria Ranieri. 2013. Is it a tool suitable for learning? A critical review of the literature on Facebook as a technology-enhanced learning environment. *Journal of Computer Assisted Learning* 29, 6 (2013), 487–504.
 - [57] Michael E McCullough, Shelley D Kilpatrick, Robert A Emmons, and David B Larson. 2001. Is gratitude a moral affect? *Psychological bulletin* 127, 2 (2001), 249.
 - [58] Monica Mendini, Paula C Peter, and Salvatore Maione. 2022. The potential positive effects of time spent on Instagram on consumers’ gratitude, altruism, and willingness to donate. *Journal of Business Research* 143 (2022), 16–26.
 - [59] Blaire Morgan, Liz Gulliford, and Kristján Kristjánsson. 2017. A new approach to measuring moral virtues: The Multi-Component Gratitude Measure. *Personality and Individual Differences* 107 (2017), 179–189.
 - [60] Bısra Müceldili, Oya Erdil, Ali Ekber Akgün, and Halit Keskin. 2015. Collective gratitude: Positive organizational scholarship perspective. *International Business Research* 8, 8 (2015), 92.
 - [61] Bonnie Nardi and Justin Harris. 2006. Strangers and friends: Collaborative play in World of Warcraft. In *Proceedings of the 2006 20th anniversary conference on Computer supported cooperative work*. 149–158.
 - [62] Catherine Nelson. 2009. Appreciating gratitude: Can gratitude be used as a psychological intervention to improve individual well-being? *Counselling Psychology Review* (2009).
 - [63] A Ant Ozok and Panayiotis Zaphiris. 2009. *Online communities and social computing*. Springer.
 - [64] Katherine Panciera, Aaron Halfaker, and Loren Terveen. 2009. Wikipedians are born, not made: a study of power editors on Wikipedia. In *Proceedings of the ACM 2009 international conference on Supporting group work*. 51–60.
 - [65] Orestis Papakyriakopoulos, Juan Carlos Medina Serrano, and Simon Hegelich. 2020. Political communication on social media: A tale of hyperactive users and bias in recommender systems. *Online Social Networks and Media* 15 (2020), 100058.
 - [66] Joon Sung Park, Joseph Seering, and Michael S Bernstein. 2022. Measuring the Prevalence of Anti-Social Behavior in Online Communities. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–29.
 - [67] Pranathi Ramachandra, Sara Booth, Thirza Pieters, Kalliopi Vrotsou, and Felicia A Huppert. 2009. A brief self-administered psychological intervention to improve well-being in patients with cancer: results from a feasibility study. *Psycho-Oncology: Journal of the Psychological, Social and Behavioral Dimensions of Cancer* 18, 12 (2009), 1323–1326.
 - [68] Juliana Raskauskas and Ann D Stoltz. 2007. Involvement in traditional and electronic bullying among adolescents. *Developmental psychology* 43, 3 (2007), 564.
 - [69] Annie Regan, Lisa C Walsh, and Sonja Lyubomirsky. 2023. Are some ways of expressing gratitude more beneficial than others? Results from a randomized controlled experiment. *Affective Science* 4, 1 (2023), 72–81.
 - [70] Tyler L Renshaw and Margaret L Hindman. 2017. Expressing gratitude via instant communication technology: A randomized controlled trial targeting college students’ mental health. *Mental Health & Prevention* 7 (2017), 37–44.
 - [71] George Ritzer and Nathan Jurgenson. 2010. Production, consumption, presumption: The nature of capitalism in the age of the digital ‘prosumer’. *Journal of consumer culture* 10, 1 (2010), 13–36.
 - [72] Robert C Roberts. 2004. The blessings of gratitude: A conceptual analysis. *The psychology of gratitude* (2004), 58–78.
 - [73] Bobby Rozzell, Cameron W Piercy, Caleb T Carr, Shawn King, Brianna L Lane, Michael Tornes, Amy Janan Johnson, and Kevin B Wright. 2014. Notification pending: Online social support from close and nonclose relational ties via Facebook. *Computers in Human Behavior* 38 (2014), 272–280.
 - [74] Mattia Samory and Enoch Peserico. 2017. Sizing up the troll: A quantitative characterization of moderator-identified trolling in an online forum. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 6943–6947.
 - [75] Martin Saveski, Brandon Roy, and Deb Roy. 2021. The structure of toxic conversations on twitter. In *Proceedings of the Web Conference 2021*. 1086–1097.
 - [76] Nathan Schneider, Primavera De Filippi, Seth Frey, Joshua Z Tan, and Amy X Zhang. 2021. Modular politics: Toward a governance layer for online communities. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–26.
 - [77] Simona Sciarra, Daniela Villani, Anna Flavia Di Natale, and Camillo Regalia. 2021. Gratitude and social media: a pilot experiment on the benefits of exposure to others’ grateful interactions on facebook. *Frontiers in Psychology* 12 (2021), 667052.
 - [78] Joseph Seering, Robert Kraut, and Laura Dabbish. 2017. Shaping pro and anti-social behavior on twitch through moderation and example-setting. In *Proceedings of the 2017 ACM conference on computer supported cooperative work and social computing*. 111–125.
 - [79] Martin EP Seligman, Tracy A Steen, Nansook Park, and Christopher Peterson. 2005. Positive psychology progress: empirical validation of interventions. *American psychologist* 60, 5 (2005), 410.
 - [80] Mina Seraj. 2012. We create, we connect, we respect, therefore we are: intellectual, social, and cultural value in online communities. *Journal of Interactive Marketing* 26, 4 (2012), 209–222.
 - [81] Prina Shachaf and Noriko Hara. 2010. Beyond vandalism: Wikipedia trolls. *Journal of Information Science* 36, 3 (2010), 357–370.
 - [82] Kennon M Sheldon and Sen-chi Yu. 2022. Methods of gratitude expression and their effects upon well-being: Texting may be just as rewarding as and less risky than face-to-face. *The Journal of Positive Psychology* 17, 5 (2022), 690–700.
 - [83] Garriy Shteynberg. 2015. Shared attention. *Perspectives on psychological science* 10, 5 (2015), 579–590.
 - [84] Adam Smith. 2010. *The theory of moral sentiments*. Penguin.
 - [85] John Suler. 2004. The online disinhibition effect. *Cyberpsychology & behavior* 7, 3 (2004), 321–326.
 - [86] Qiao Tang, Xinmu Hu, Ziwen Zeng, and Yufeng Zhao. 2022. Co-Orb: Fostering Remote Workplace Gratitude with IoT Technology. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–6.
 - [87] Jo-Ann Tsang. 2021. (Un) special Favors: Gratitude for Group-Based Benefits. *The Journal of Positive Psychology* 16, 1 (2021), 27–37.
 - [88] Sonja Utz. 2009. ‘Egoboo’ vs. altruism: the role of reputation in online consumer communities. *New Media & Society* 11, 3 (2009), 357–374.
 - [89] Sonja Utz and Johannes Breuer. 2017. The relationship between use of social network sites, online social support, and well-being. *Journal of media psychology* (2017).
 - [90] Ahmad Valikhani, Fatemeh Ahmadnia, Alma Karimi, and Paul J Mills. 2019. The relationship between dispositional gratitude and quality of life: The mediating role of perceived stress and mental health. *Personality and Individual Differences* 141 (2019), 40–46.
 - [91] Kris Varjas, Jasmine Talley, Joel Meyers, Leandra Parris, and Hayley Cutts. 2010. High school students’ perceptions of motivations for cyberbullying: An exploratory study. *Western Journal of Emergency Medicine* 11, 3 (2010), 269.
 - [92] Jessica Vitak, Kalyani Chadha, Linda Steiner, and Zahra Ashktorab. 2017. Identifying women’s experiences with and strategies for mitigating negative effects of online harassment. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 1231–1245.
 - [93] Emily A Vogels. 2021. The state of online harassment. *Pew Research Center* 13 (2021), 625.
 - [94] Lisa C Walsh, Annie Regan, and Sonja Lyubomirsky. 2022. The role of actors, targets, and witnesses: Examining gratitude exchanges in a social context. *The Journal of Positive Psychology* 17, 2 (2022), 233–249.
 - [95] Philip C Watkins, Kathrane Woodward, Tamara Stone, and Russell L Kolts. 2003. Gratitude and happiness: Development of a measure of gratitude, and relationships with subjective well-being. *Social Behavior and Personality: an international journal* 31, 5 (2003), 431–451.
 - [96] Eiswein Tsz Kin Wong, WWK Ma, and Wendy Wing Lam Chan. 2017. Exploring the relationships of online social ties, altruism, and online social support. *Journal of Communication and Education* 4, 1 (2017), 4–18.
 - [97] Alex M Wood, Jeffrey J Froh, and Adam WA Geraghty. 2010. Gratitude and well-being: A review and theoretical integration. *Clinical psychology review* 30, 7 (2010), 890–905.
 - [98] Xi Xiong, Yuanyuan Li, Shaojie Qiao, Nan Han, Yue Wu, Jing Peng, and Binyong Li. 2018. An emotional contagion model for heterogeneous social media with multiple behaviors. *Physica A: Statistical Mechanics and its Applications* 490 (2018), 185–202.
 - [99] Jun-ichi Yamamoto, Tomohiro Fukui, Kazutomo Nishii, Ichiro Kato, and Quang Thahn Pham. 2022. Digitalizing Gratitude and Building Trust through Technology in a Post-COVID-19 World—Report of a Case from Japan. *Journal of Open Innovation: Technology, Market, and Complexity* 8, 1 (2022), 22.
 - [100] Diyi Yang, Zheng Yao, Joseph Seering, and Robert Kraut. 2019. The channel matters: Self-disclosure, reciprocity and social support in online cancer support groups. In *Proceedings of the 2019 chi conference on human factors in computing systems*. 1–15.
 - [101] Mengfan Yao, Charalampos Chelmiss, and Daphney-Stavroula Zois. 2019. Cyberbullying ends here: Towards robust detection of cyberbullying in social media.

- In *The World Wide Web Conference*. 3427–3433.
- [102] Zheng Yao, Diyi Yang, John M Levine, Carissa A Low, Tenbroeck Smith, Haiyi Zhu, and Robert E Kraut. 2021. Join, stay or go? A closer look at members' life cycles in online health communities. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–22.
 - [103] Khushnood Z. Naqshbandi, Silas Taylor, Ajit G. Pillai, and Naseem Ahmadpour. 2021. Labour of love: Volunteer perceptions on building relatedness in online volunteering communities. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–6.
 - [104] Michal Zechariah. 2010. True gratitude is a communal emotion, not a wellness practice.
 - [105] Amy X Zhang, Grant Hugh, and Michael S Bernstein. 2020. PolicyKit: building governance in online communities. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. 365–378.

Received 11 May 2023; accepted 15 June 2023